Adaptive API Gateway Challenge (Student Guide)

Context

Server A recently moved from Version 1 (V1) to Version 2 (V2). The participant server is expected to help bridge the old and new models while also reporting service-health metrics from heartbeat data.

Goal

Implement a server that exposes `POST /solve` and returns a combined response based on the incoming request.

Required Endpoint

- `POST /solve`

Sample Request

The endpoint receives:

```
{
    "payload": "ewoJImFkYXB0SW5wdXQiOiB7CgkJInVzZXIiOiB7CgkJCSJpZCI6ICJV
}
```

where the payload somehow decodes to this:

```
{
    "adaptInput": {
        "user": {
            "id": "U42",
            "fullName": "Jane Doe"
        },
```

```
        "action": "CREATE",
        "metadata": {
            "priority": "HIGH"
        }
    },
    "heartbeats": [
        {
            "service": "auth",
            "timestamp": 1710000123,
            "latencyMs": 120,
            "status": "OK"
        },
        {
            "service": "auth",
            "timestamp": 1710000125,
            "latencyMs": 180,
            "status": "FAIL"
        },
        {
            "service": "auth",
            "timestamp": 1710000121,
            "latencyMs": 95,
            "status": "OK"
        }
    ],
    "sloQuery": {
        "service": "auth",
        "since": 1710000123
    }
}
```


Sample Response

Your `POST /solve` must return JSON in this shape:


```
{
    "adaptOutput": {
        "id": "U42",
        "name": "Jane Doe",
        "action": "create",
        "priority": 3
    },
    "sloOutput": {
```

```
            "availability": 0.5,
            "p95LatencyMs": 180
        }
    }
```